

Python Lab: Strings

A set of practice tasks covering string basics, manipulation, slicing, and word/sentence operations

1. Basics

- 1 Take a string input from the user and print its length.
- 2 Reverse a given string (without using slicing, then with slicing).
- 3 Check if a given string is a palindrome.
- 4 Convert a string to uppercase and lowercase.
- 5 Count the number of vowels, consonants, digits, and spaces in a string.

2. Manipulation

- 6 Count the occurrences of a specific character in a string.
- 7 Remove all whitespace from a string.
- 8 Replace all occurrences of a character/word with another.
- 9 Concatenate two strings without using the + operator.
- 10 Swap the case of each character in a string (upper → lower and vice versa).

3. Slicing & Indexing

- 11 Extract the first and last characters of a string.
- 12 Print every second character of a string.
- 13 Check if a substring exists within a string.
- 14 Find the index of the first and last occurrence of a character.

4. Words & Sentences

- 15 Count the number of words in a sentence.
- 16 Find the longest word in a sentence.
- 17 Reverse the order of words in a sentence (without reversing each word).
- 18 Capitalize the first letter of every word (Title Case), without using .title().
- 19 Check if two strings are anagrams of each other.

5. Advanced

- 20 Remove duplicate characters from a string.

- 21 Check if a string contains only digits, only alphabets, or is alphanumeric.
- 22 Find all the duplicate characters in a string and their counts.
- 23 Convert a string into a list of characters and back into a string.
- 24 Write a program to check if a string is a valid identifier (like a Python variable name).
- 25 Implement your own version of `str.find()` and `str.count()` without using built-in string methods.